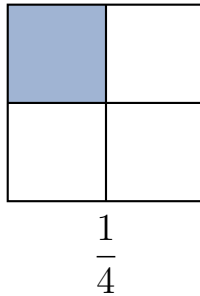


Fractions: Introduction

Introduction to Fractions

A **fraction** is an *equal* part of a whole. The square below has been divided into 4 equal parts. The fraction that represents the shaded part of the square is $\frac{1}{4}$, since 1 of the 4 parts is shaded.



Alt text: A square divided into four equal quadrants by a horizontal and a vertical line through its center. The top-left quadrant is shaded blue, representing the fraction one-fourth.

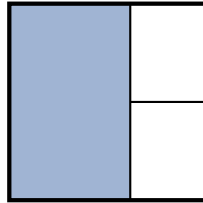
The **denominator** of a fraction tells you how many equal-sized pieces a whole has been divided into. The **numerator** of a fraction is how many of those pieces you are counting or considering.

Why do we need to divide a whole into equal parts?

Because fractions only make sense if the whole is divided into equal parts. For example, the INCORRECT square on the left has been divided into 3 parts and 1 of them is shaded. However, we can't say that $\frac{1}{3}$ of the square is shaded because the parts are not the same size. To correct this, we redraw the lines so that every part is the exact same size, as shown in the CORRECT square on the right.

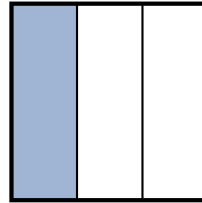
Author(s): Kelly Stady

INCORRECT



$\frac{1}{3}$?

CORRECT

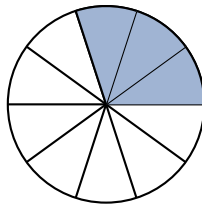


$\frac{1}{3}$

Alt text: Two squares side-by-side demonstrating the importance of equal parts in fractions. The left square is labeled INCORRECT with a red X; it is divided into three unequal parts with one large section shaded, showing that counting unequal pieces as one-third is incorrect.

The right square is labeled CORRECT with a green checkmark; it is divided into three identical vertical rectangles with one shaded, correctly representing the fraction one-third.

Problem 1 The circle below has been divided into 10 equal parts.

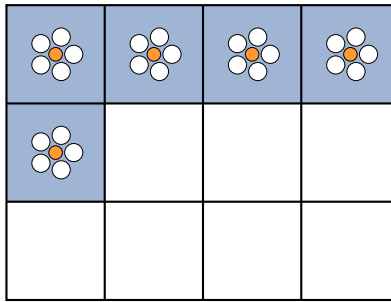


Alt text: A circle representing a whole, divided into 10 equal pie-shaped wedges. Three adjacent wedges in the top-right section are shaded blue.

What fraction of the circle is shaded?

Fraction Shaded = $\frac{\dots\dots}{\dots\dots}$

Problem 2 A local wildlife sanctuary covers 12 acres of protected land. Recent data shows that 5 acres have been successfully restored with native wild flowers to support declining bee populations.



Each box represents 1 acre of the sanctuary.

Alt text: A grid of 12 equal squares arranged in three rows of four. The four squares in the top row and the first square in the middle row are shaded blue with white flowers in the center.

What fraction of the sanctuary's total area has been restored with wild flowers?

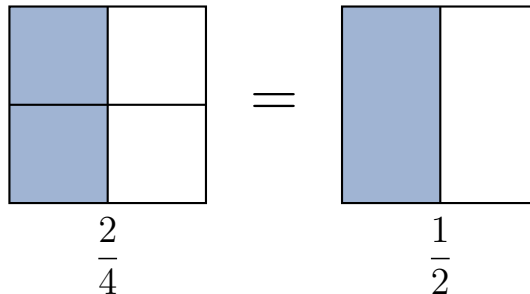
.....

What fraction of the sanctuary's total area has **not** been restored with wild flowers?

.....

Equivalent Fractions

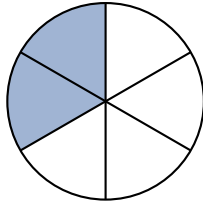
The square on the left has been divided into 4 equal parts, with 2 parts shaded. The square on the right has been divided into 2 equal parts, with 1 part shaded. Since both squares are the same size and have an equal amount of shading, we can conclude that $\frac{2}{4}$ is equivalent to $\frac{1}{2}$.



Alt text: Two squares side-by-side, as described above, separated by an equal sign. The square on the left represents two-fourths and the square on the right represents one-half.

Equivalent fractions represent the same part of a whole, but have different numerators and denominators. Since a whole can be divided into any number of equal parts, each fraction has infinitely many equivalent fractions.

Problem 3 Consider the circle below.



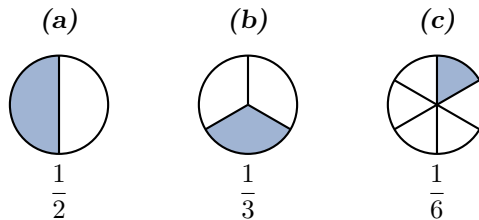
Alt text: A circle divided into six equal wedge-shaped sections. Two adjacent sections on the upper-left are shaded blue.

What fraction of the circle is shaded?

$$\text{Fraction Shaded} = \frac{\dots\dots}{\dots\dots}$$

Which of the following circles represents an **equivalent** fraction?

((a)/ (b)/ (c))



Alt text: Three circles labeled (a), (b), and (c) representing fractions. Circle (a) is divided into 2 equal parts with 1 shaded (one-half). Circle (b) is divided into 3 equal parts with 1 shaded (one-third). Circle (c) is divided into 6 equal parts with 1 shaded (one-sixth).

Definition 1 (Types of Fractions). (a) **Proper fractions** are fractions in which the numerator is smaller than the denominator. A proper fraction is always less than 1.

• **Example:** $\frac{3}{4}$

(b) **Improper fractions** are fractions in which the numerator is greater than or equal to the denominator. An improper fraction is always greater than or equal to 1.

- **Example:** $\frac{7}{12}$

(c) **Mixed numbers** consist of a whole number and a fraction. A mixed number is always greater than 1.

- **Example:** $1\frac{1}{4}$

Problem 4 Classify each of the following fractions as a **proper fraction**, an **improper fraction** or a **mixed number**.

(a) $2\frac{1}{2}$

Multiple Choice:

- (i) *Proper Fraction*
- (ii) *Improper Fraction*
- (iii) *Mixed Number*

(b) $\frac{11}{12}$

Multiple Choice:

- (i) *Proper Fraction*
- (ii) *Improper Fraction*
- (iii) *Mixed Number*

(c) $\frac{15}{8}$

Multiple Choice:

- (i) *Proper Fraction*
- (ii) *Improper Fraction*
- (iii) *Mixed Number*

(d) $\frac{5}{5}$

Multiple Choice:

- (i) *Proper Fraction*
- (ii) *Improper Fraction*
- (iii) *Mixed Number*

Problem 5 When we write the mixed number $1\frac{1}{4}$ what arithmetic symbol is not written, but understood to be between the 1 and $\frac{1}{4}$?

Multiple Choice:

- (a) $\times$
- (b) $-$
- (c) $+$
- (d) $\div$

Negative Fractions

In the following problems, notice that the position of the negative sign changes but the value of the fraction remains the same.

$$\frac{-4}{2} = -2$$

$$\frac{4}{-2} = -2$$

$$-\frac{4}{2} = -2$$

Therefore, a negative sign in a fraction can be written in the numerator, the denominator or in front of the fraction. However, when simplifying we typically place the negative sign in front of the fraction.